Intelligent Study Material Query System using RAG

AI-Powered Conversational Learning Platform

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QUESTION-7.2

Institution

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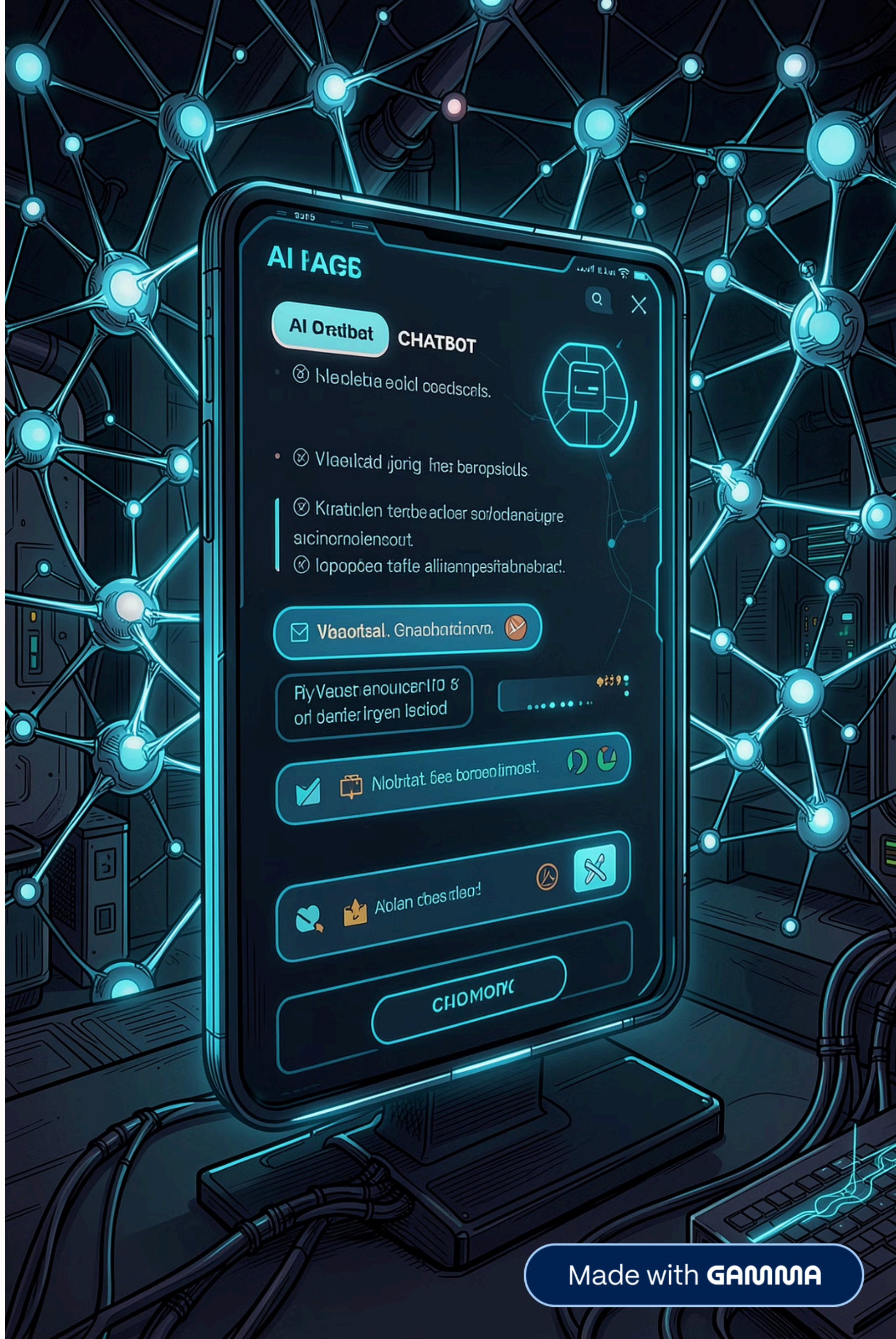


Problem Statement & Objectives

The Problem

- Manually searching through PDFs and DOCX files is time-consuming and inefficient
- Students struggle to locate precise answers buried in lengthy documents
- Traditional keyword search lacks semantic understanding and context
- No conversational interface exists to make study material interactive

Our Objectives

Intelligent Document Querying

Natural language Q&A over uploaded study material

PDF & DOCX Ingestion

Seamless multi-format document support

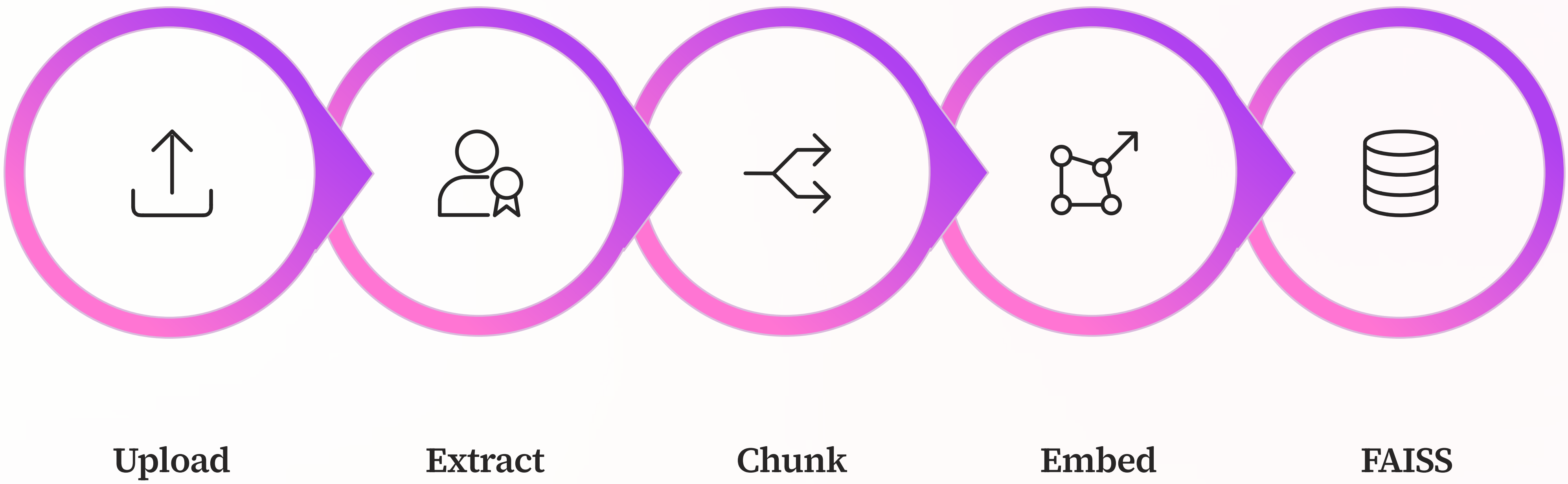
Semantic Search

Embedding-based similarity retrieval via FAISS

Context-Aware Answers

LLM generates precise, grounded responses

System Architecture & RAG Workflow



The pipeline transforms raw study documents into a queryable semantic knowledge base, enabling precise, context-aware answers via a large language model.

Text Splitter

```
splitter = RecursiveCharacterTextSplitter(
  chunk_size=200,
  chunk_overlap=50
)
```

Embedding Model

```
embedding_model = HuggingFaceEmbeddings(
  model_name="sentence-transformers/
  all-MiniLM-L6-v2"
)
```

Vector Store Creation

```
vectorstore = FAISS.from_documents(
  chunks,
  embedding_model
)
```

Chunk size experiments were conducted at **200 tokens** and **500 tokens** to evaluate retrieval trade-offs.

Comparison & Experimental Results

Parameter	Chunk 200	Chunk 500
Retrieval Precision	High	Medium
Context Coverage	Low	High
Retrieval Noise	Low	Medium
Answer Detail	Concise	Detailed
Processing Speed	Faster	Slower
Best Use Case	Fact Q&A	Conceptual

```
docs = vectorstore.similarity_search(
    question, k=3
)
```

1

Smaller Chunks

Higher precision; ideal for specific factual queries

2

Larger Chunks

Richer context; better for conceptual understanding

3

Hybrid Chunking

Balanced precision and context for general use

Conclusion & Future Scope

Key Conclusions

- **RAG significantly enhances** educational search by grounding answers in actual document content
- **FAISS enables** fast, scalable semantic retrieval across large document corpora
- **Chunk size** is a critical hyperparameter directly influencing answer quality and precision

Future Scope

Hybrid Chunking

Voice AI Tutor

Personalised Learning

Multimodal RAG

Cloud Deployment